RepertIA

Illustrated guide

Version 2.20.2 · Real screenshots of the program, made with acceptance-test data. Names and operator are fictional.

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1. What RepertIA is

RepertIA is a forensic analysis tool for Mac (Apple Silicon), Windows and Linux (experimental), with an interface in Italian or English. It takes a forensic copy of a disk or a smartphone extraction, indexes it with recognised forensic tools and lets you examine it on your own computer, with or without artificial intelligence.

What it can do

- **Mass storage:** raw, dd, img, E01 and dmg copies of disks, SSDs, pendrives and cards. File system tree, deleted files, unallocated space, timeline, document text, photo and video metadata, mail, Windows, Mac and Linux artifacts.
- **Smartphones:** full file system extractions of iPhones and Android phones (Cellebrite, GrayKey and others, as zip, tar, tar.gz or folder). Chats, calls, address books and hundreds of artifacts read with iLEAPP and ALEAPP.
- **WhatsApp:** own readers for iPhone and Android, with message types, read states and attachments; decryption of Android `.crypt14` backups when the key is in the extraction.
- **iOS sysdiagnose and keychain:** acquired separately, recorded with their SHA-256 and read with iLEAPP.
- **Search:** by words, regular expressions or hash, down to the raw bytes of the whole copy.
- **Checks:** copy hash, hash-chained operations log, hash lists, cross-check with another tool.
- **Reports:** PDF, Word and a portable report for people without the forensic copy.
- **Optional AI agent:** a local model answers questions using read-only tools and citing its sources.

Two principles

- **Everything stays on your computer.** No accounts, cloud or telemetry. The program contacts no website during analysis.
- **The forensic copy is never changed.** It is read-only and its SHA-256 can be re-verified at any time.

> **Limits to keep in mind.** RepertIA works without AI too. When used, AI can be wrong: each of its findings stays "to be checked" until the analyst confirms it. Before using it on a case, run the acceptance test on your computer and keep the report. The Linux version is experimental. Complete sysdiagnose archives and real GrayKey or Cellebrite keychains have not been tested.

> **The data in this guide.** The screenshots are real, but all cases use acceptance-test data: public or constructed images with fictional names. The keychain and the WhatsApp backup are **fake**, with invented keys. Some sections therefore look almost empty: on a real copy they will hold far more results.

2. Installing and starting

Mac (Apple Silicon)

Unzip `RepertIA-2.20.2-Mac.zip` and double-click the app. The first time macOS may block it: right-click, **Open**, then **Open** again. On first launch it installs any missing tools by itself, using Homebrew and the internet. Keep the Terminal window open while you work.

Windows 10 and 11 (64-bit)

Run `RepertIA-2.20.2-Setup.exe`: it installs for the current user, without administrator rights. Start it from the Start menu. Alternatively use the portable zip.

Linux x86_64 (experimental)

Extract `RepertIA-2.20.2-Linux.tar.gz`, then run `./installa.sh` and `./RepertIA.sh`.

Requirements

- **Memory:** 8 GB is enough for pendrives and small disks; 16 GB is comfortable.
- **Disk:** preferably an SSD. A case is not a copy of the disk but a database of indexes.
- **AI model (optional):** a model with *tool calling* and a context of at least 16k tokens, preferably 32k.

Connecting the AI model

Open LM Studio, load a model with a large context and start the server from the **Developer** tab. RepertIA finds it by itself. llama-server and Ollama work too. Without a server the program still works: all data stays available.

Acceptance test

Download `RepertIA-collaudo.zip`, extract it next to the program and run the test: on Mac `Collaudo RepertIA.command`, on Windows from the Start menu, on Linux `collaudo.sh`. Keep the `collaudo_<date>.md` report with your expert report.

3. Mass storage: your first case

This chapter follows a complete case on a small test pendrive, with three photos, one of them deleted, and a text file.

Start page

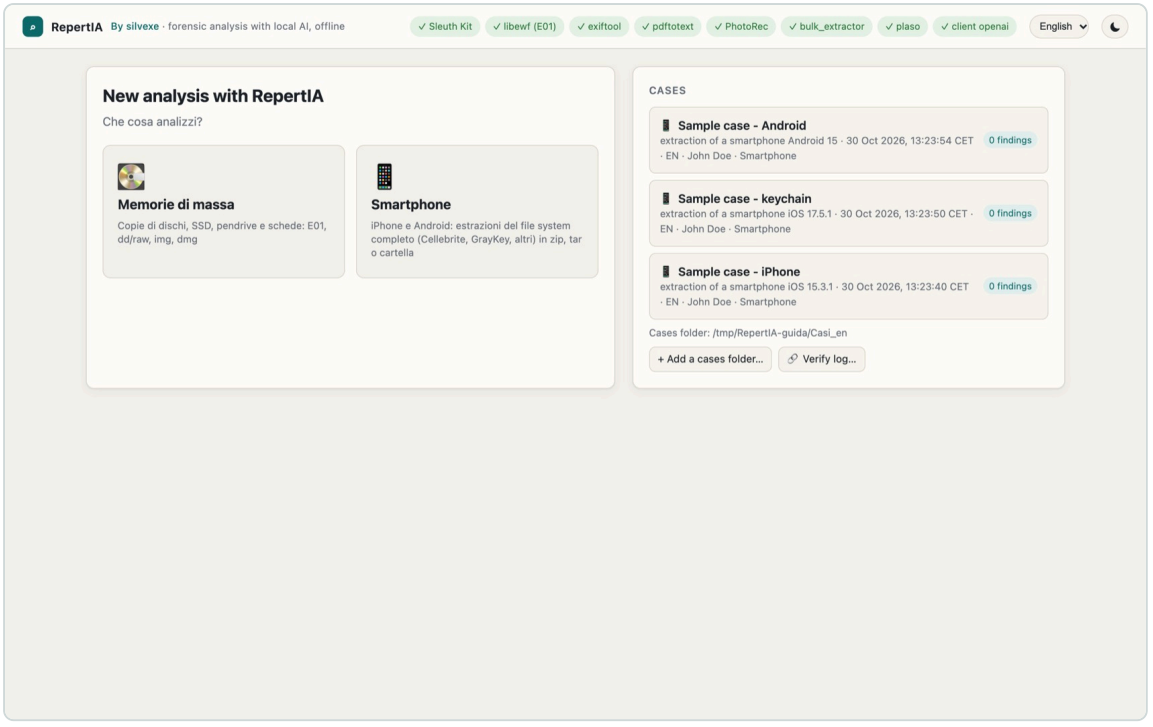


Figure 1. Here you choose what to analyse: **Mass storage** or **Smartphone**. Existing cases are listed below: one click reopens them without redoing anything.

Choosing the forensic copy

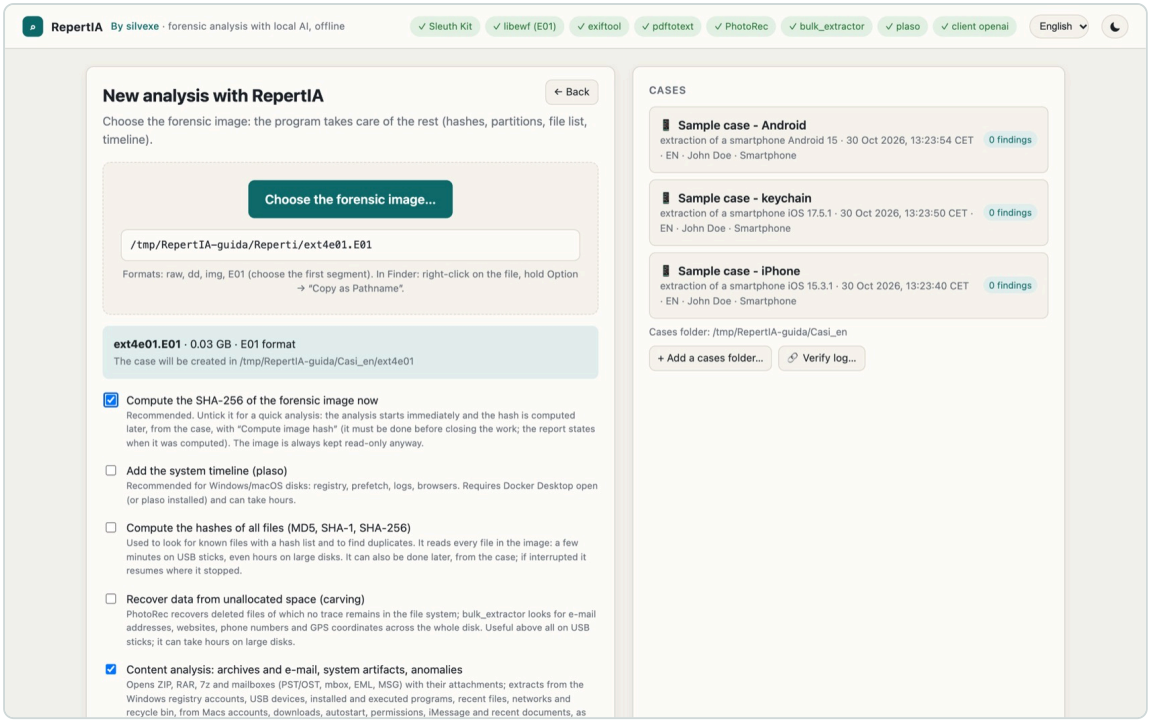


Figure 2. Press the button to choose the forensic copy, or paste its path. For E01 files choose the first segment. Computing the SHA-256 is recommended; turn it off only for a quick analysis, and compute it before closing the work.

Case details

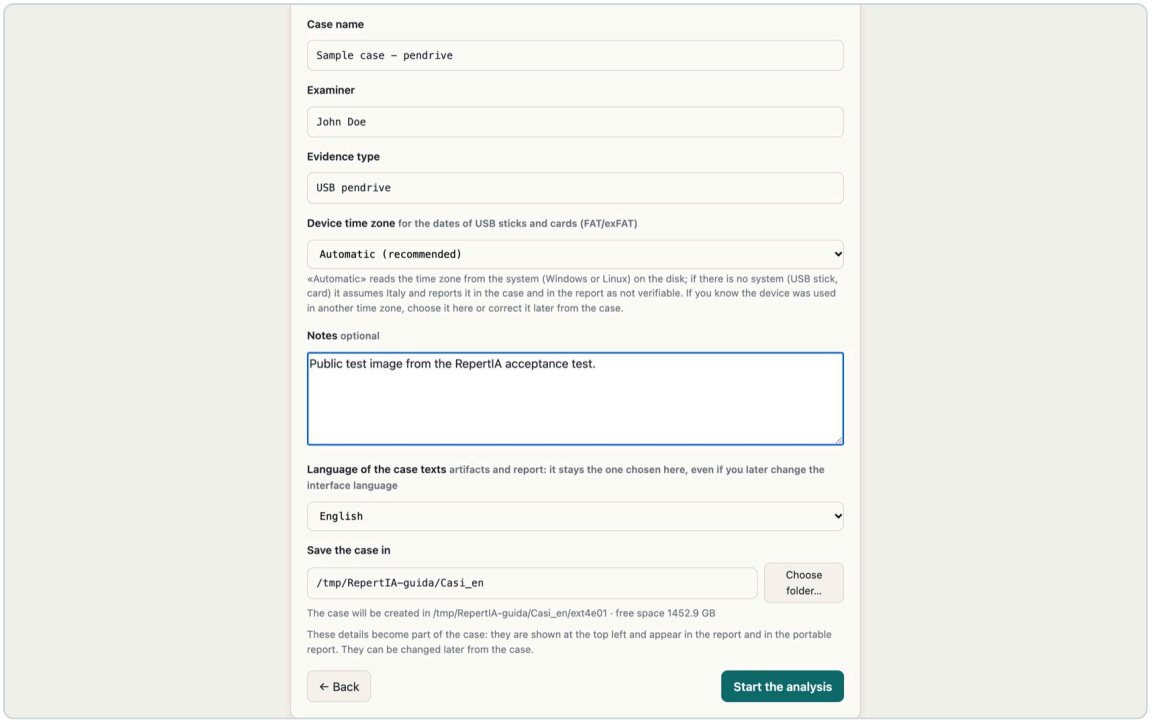


Figure 3. Case name, operator, exhibit and notes are optional, but they go into the report. John Doe is a fictional name.

Case preparation

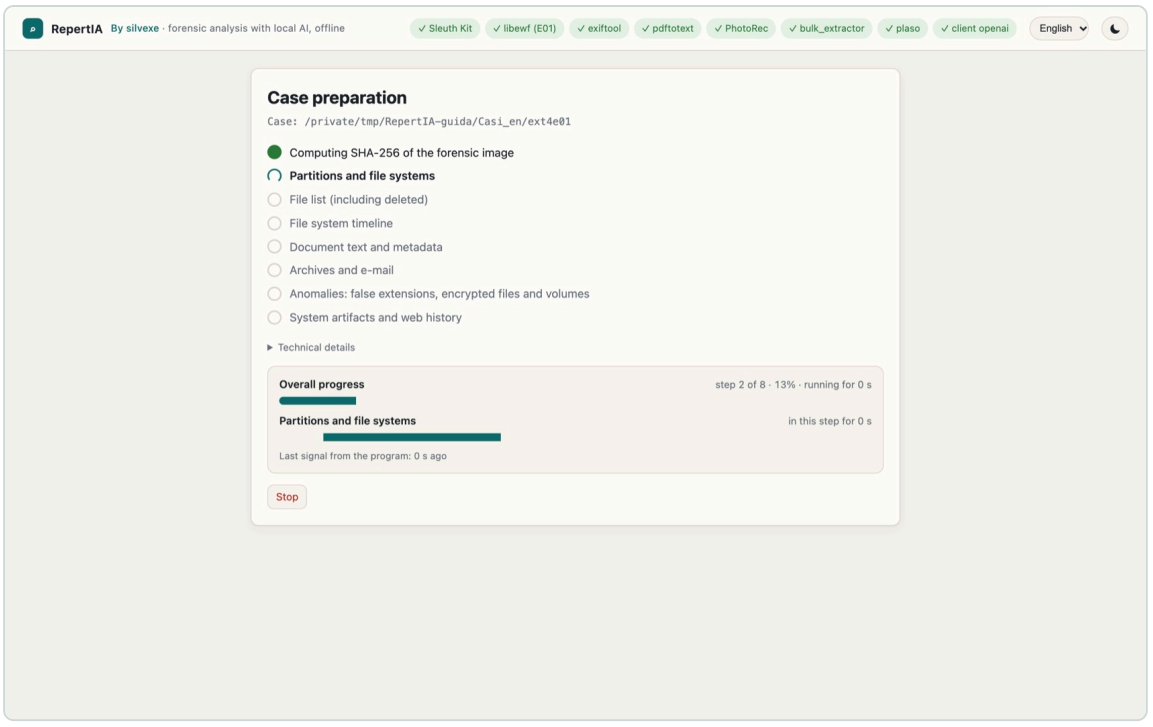


Figure 4. RepertIA works through the steps in turn: hash, partitions, file list, timeline, text, metadata, archives, anomalies and artifacts. Green dots mark finished steps.

The case is ready

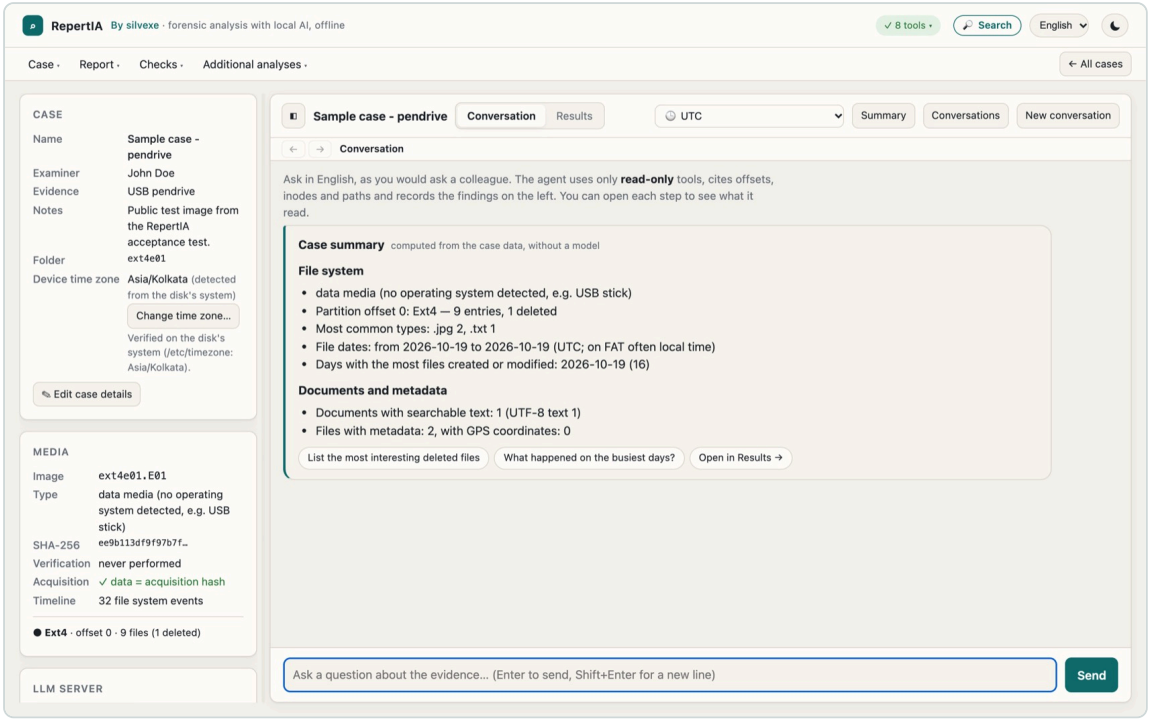


Figure 5. On the left, case and media details, including the hash result. On the right, the **case summary**, computed from the data without an AI model. The selector at the top changes the time zone used to display times.

The Results tab

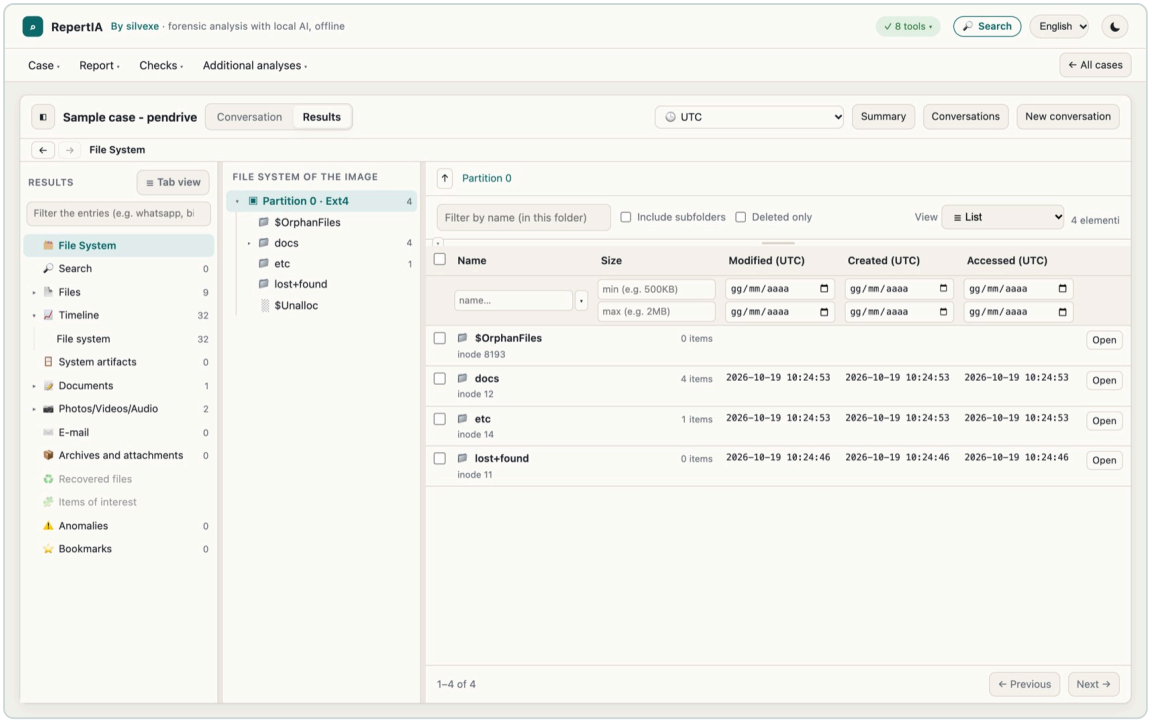


Figure 6. All data can be browsed here, even without AI. Sections on the left, tree in the middle, contents on the right. **$OrphanFiles** holds files without a folder, **$Unalloc** the unallocated space.

Opening a folder

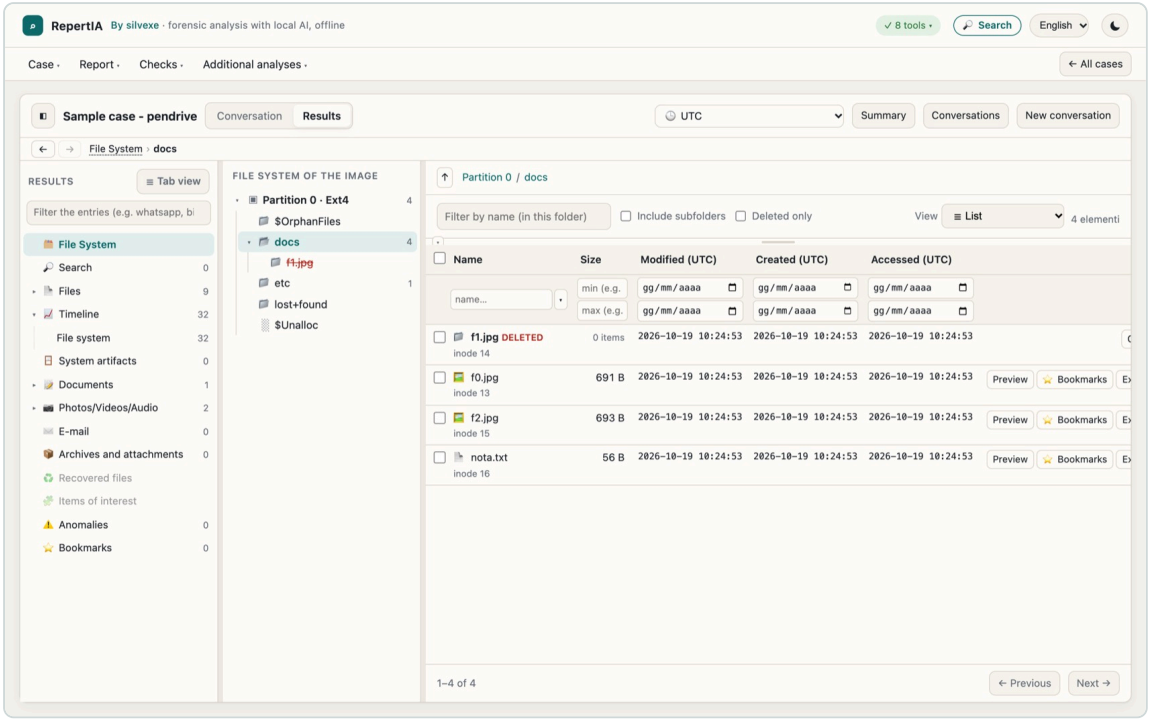


Figure 7. A click on a folder shows its contents, with size, inode and dates. Deleted entries are red. Here `f1.jpg` has a folder icon because its inode 14 was reassigned to the `/etc` folder.

Previewing a file

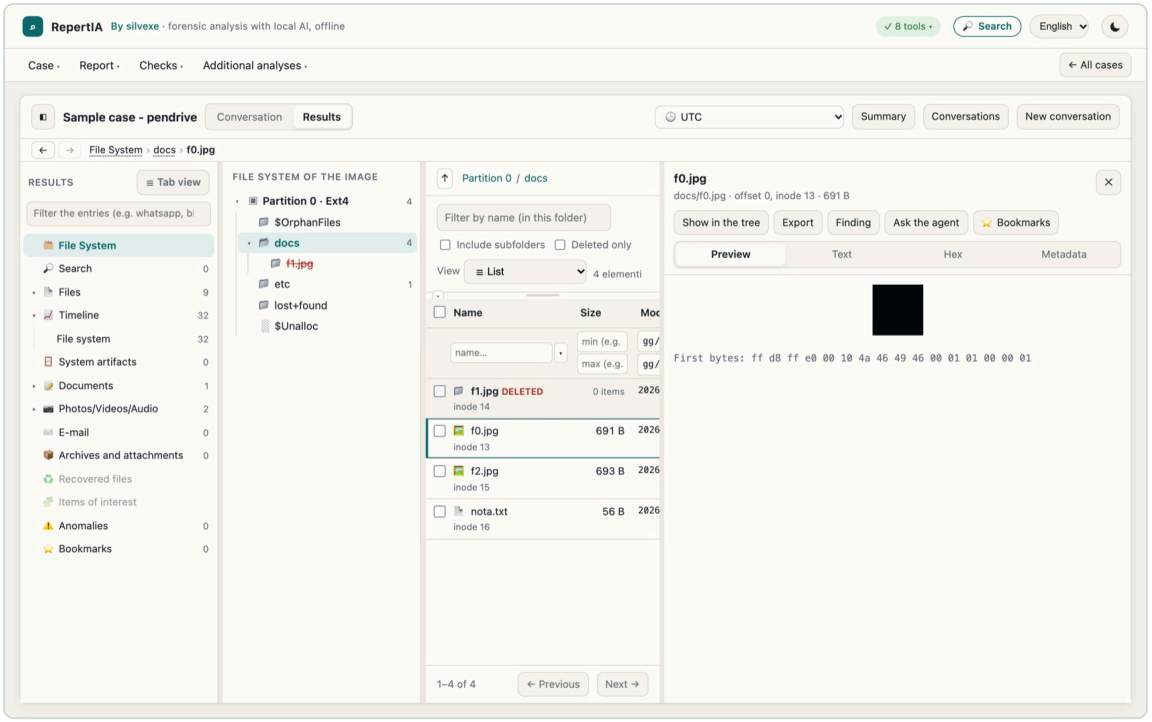


Figure 8. A click on a file opens four tabs: **Preview**, **Text**, **Hex** and **Metadata**. The buttons locate it in the tree, export it, record it as a finding, pass it to the agent or add it to Bookmarks. The dark square is the real content of this test file.

Deleted files only

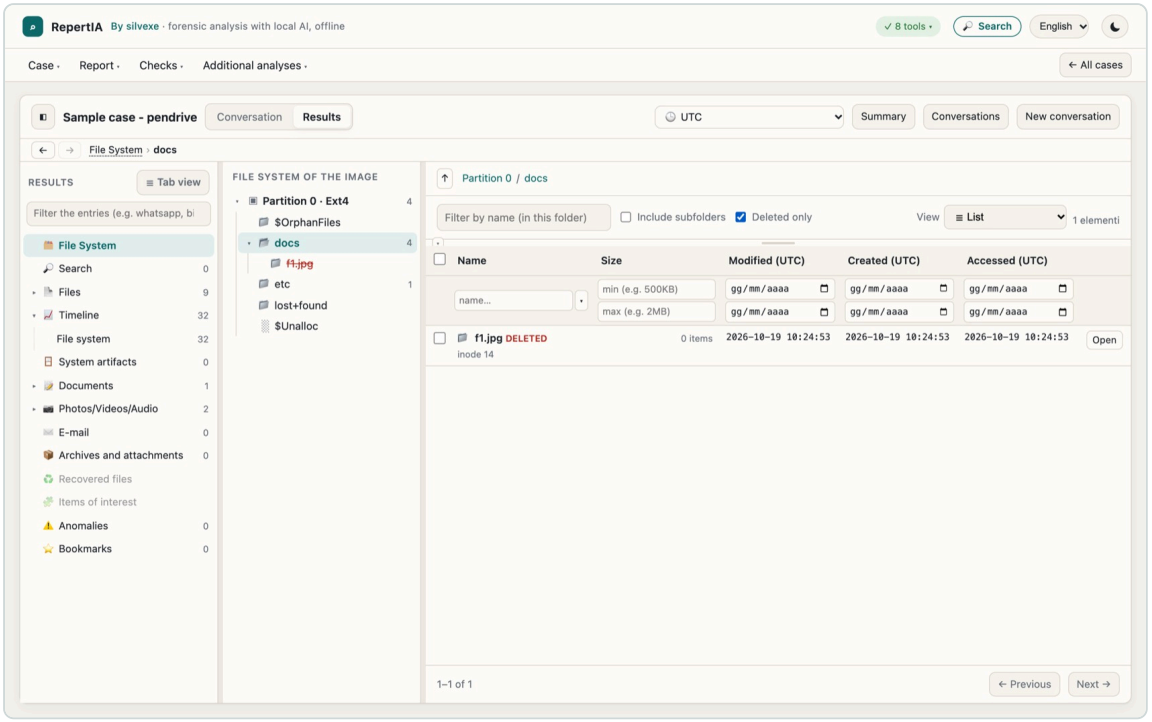


Figure 9. The box filters the open folder; with **Include subfolders** it extends to those inside.

Timeline

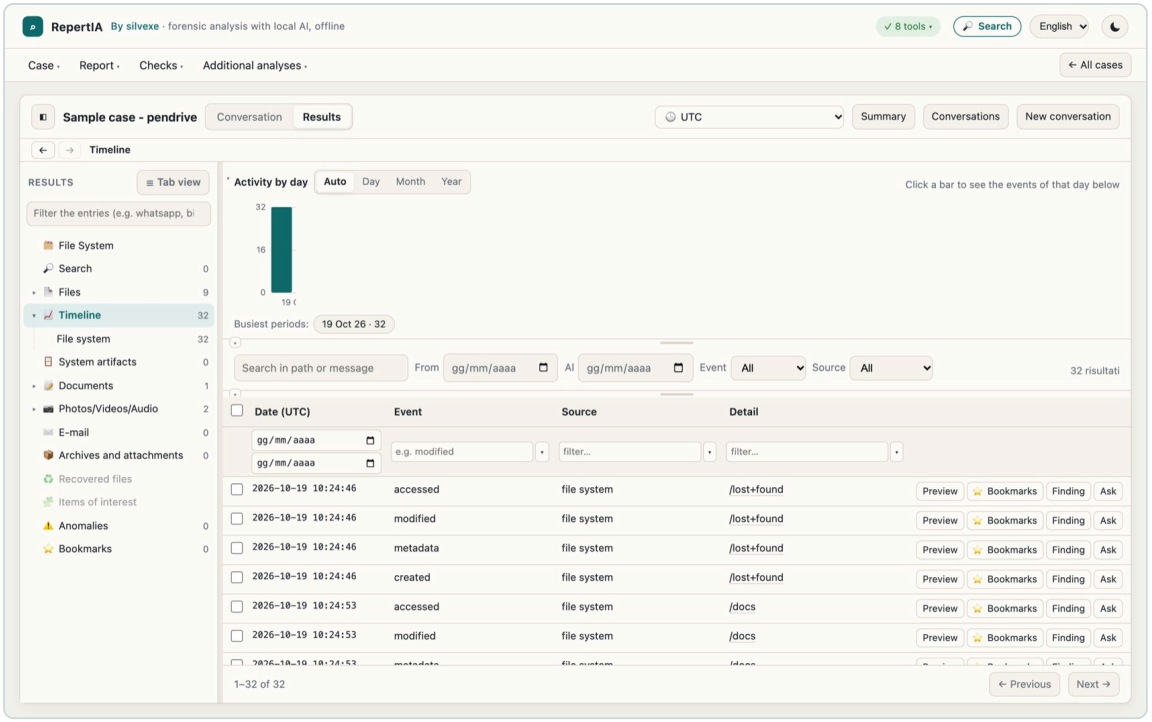


Figure 10. The chart shows activity over time: clicking a bar filters that day. The table lists modification, access, metadata and creation, with filters by date, event and source.

Search

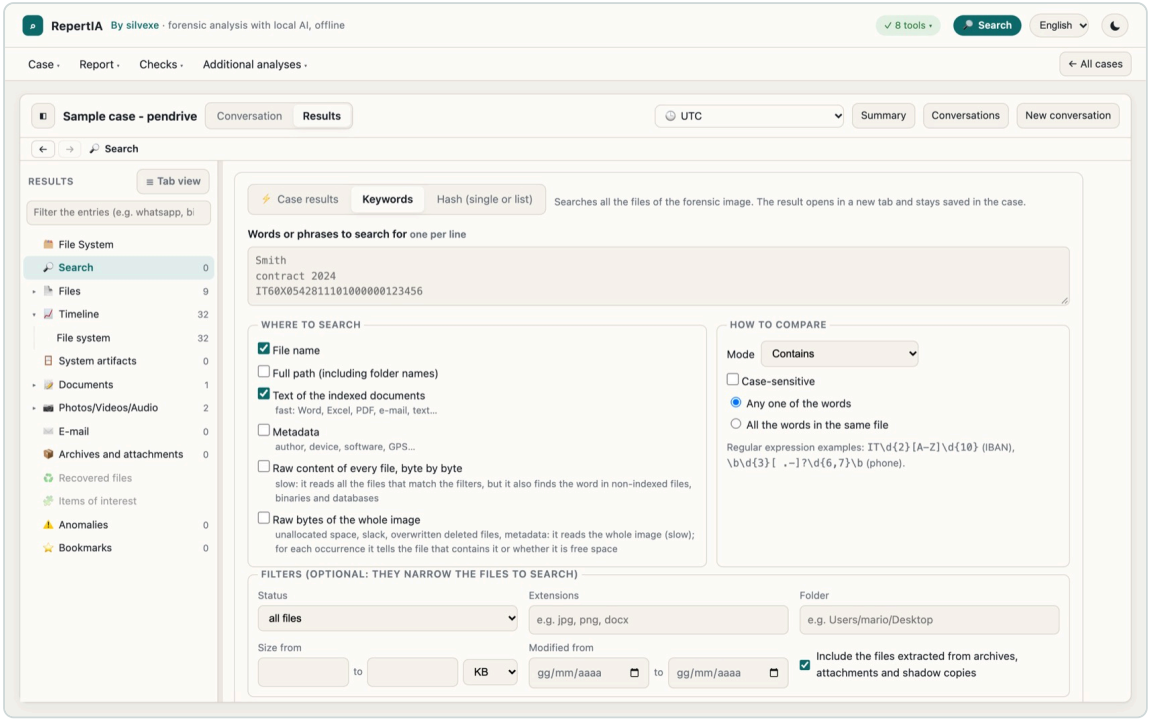


Figure 11. Enter words or phrases, one per line, or use the **Hash** tab. Choose where to search, from file names to the raw bytes of the whole copy. The result opens in its own tab and stays saved in the case.

The Report menu

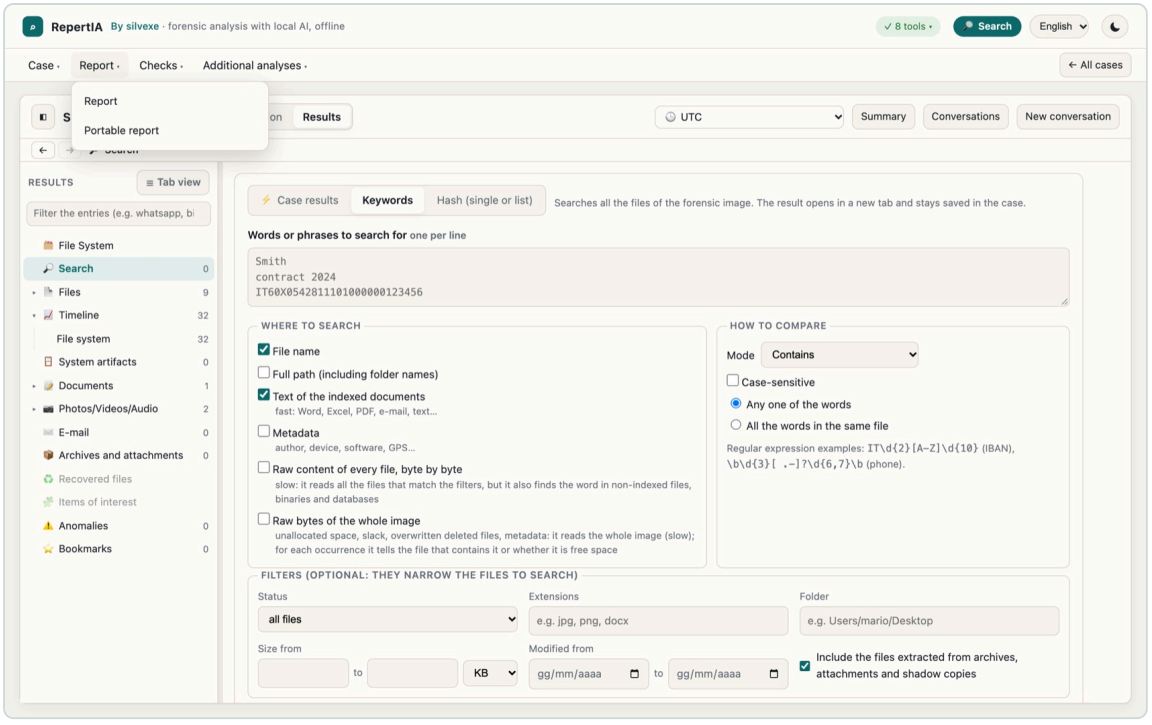


Figure 12. This is where you create the case report and the portable report.

The case report

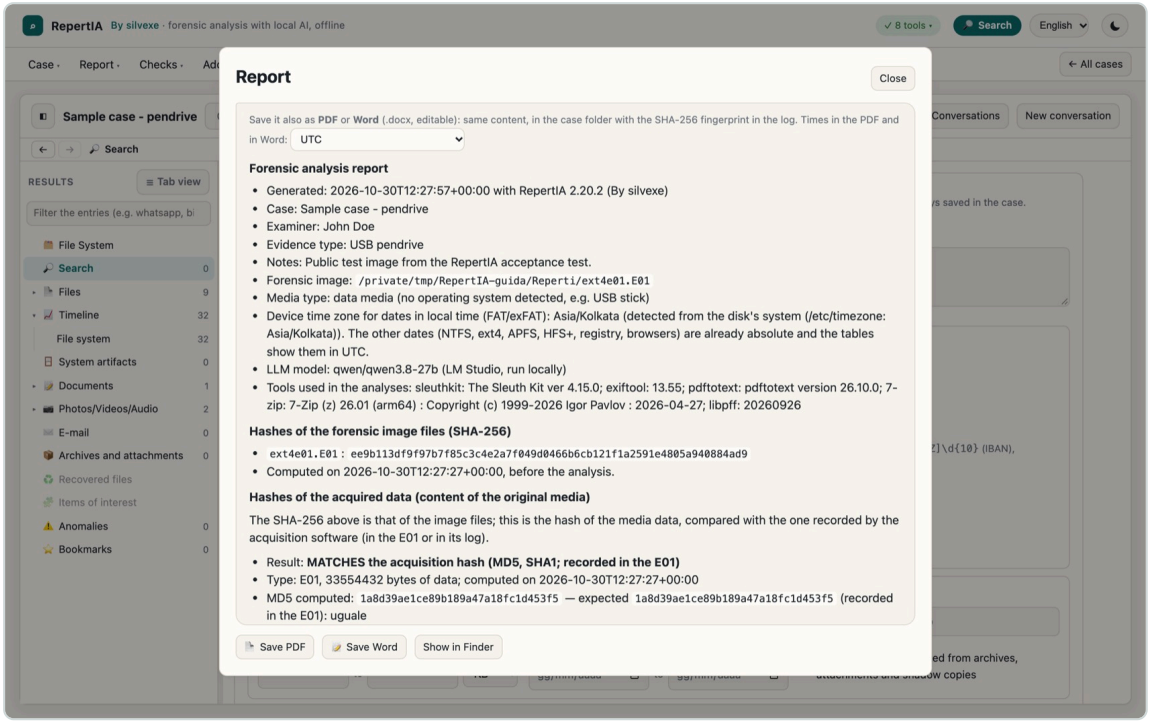


Figure 13. The report can be read in the window and saved as **PDF** or **Word**. It also covers the operations log, with the reference for checking it later.

4. Additional analyses, checks and portable report

With a case open, the bar at the top holds the other functions.

Additional analyses

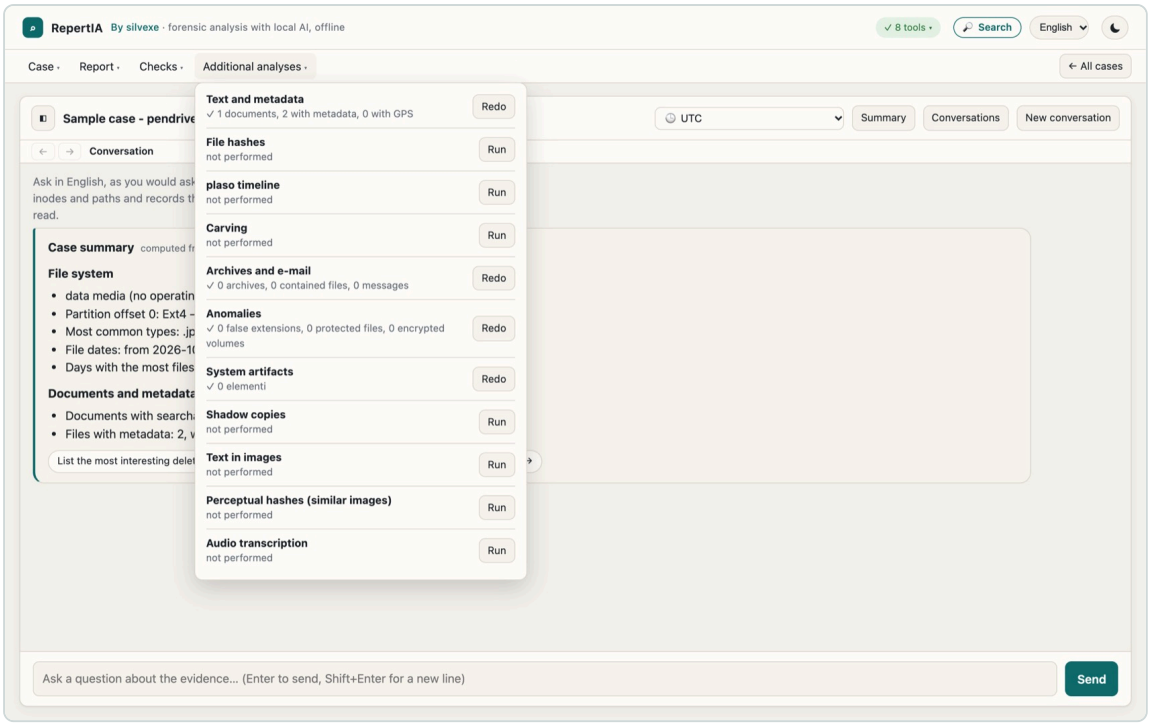


Figure 14. Each analysis runs when you need it and shows its state. **File hashes** computes MD5, SHA-1 and SHA-256 of every file. **plaso timeline** adds registry, prefetch, logs and browsers, and can take hours. **Carving** recovers files from free space with PhotoRec and looks for traces with bulk_extractor. There are also shadow copies, text in images, perceptual hashes and audio transcription.

Checks

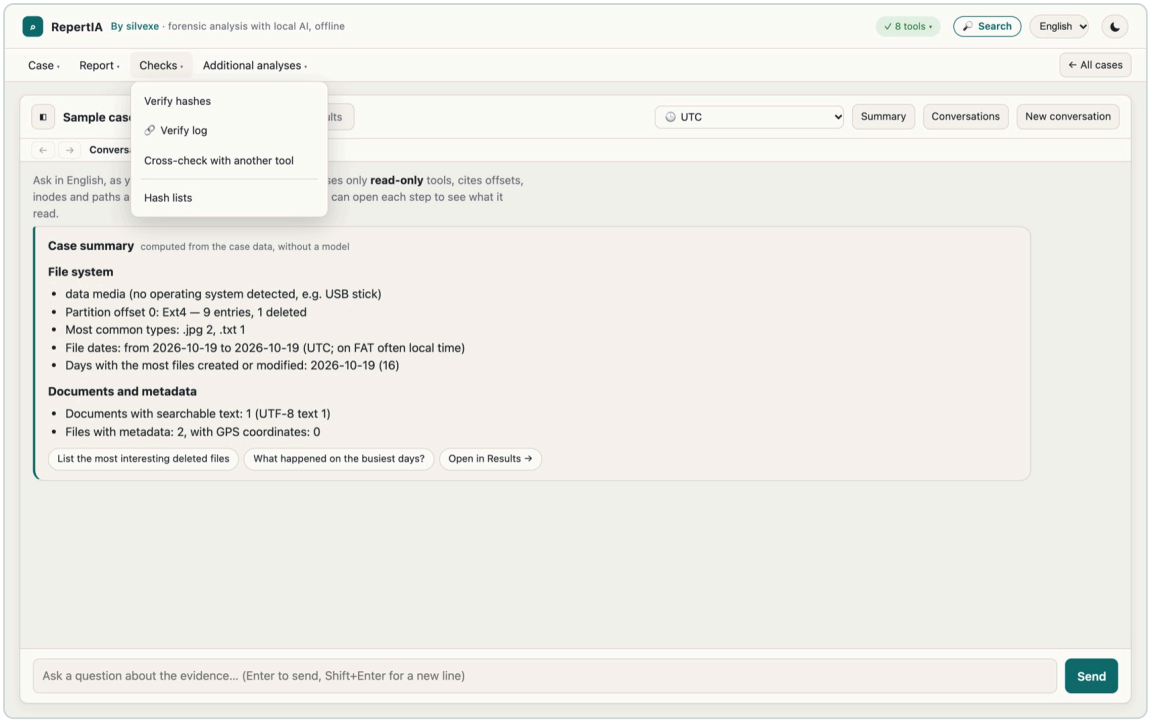


Figure 15. Verify hashes re-checks the forensic copy. **Verify log** checks that the hash-chained log is intact. **Cross-check with another tool** compares results with an export from another program. **Hash lists** looks for known files.

Portable report

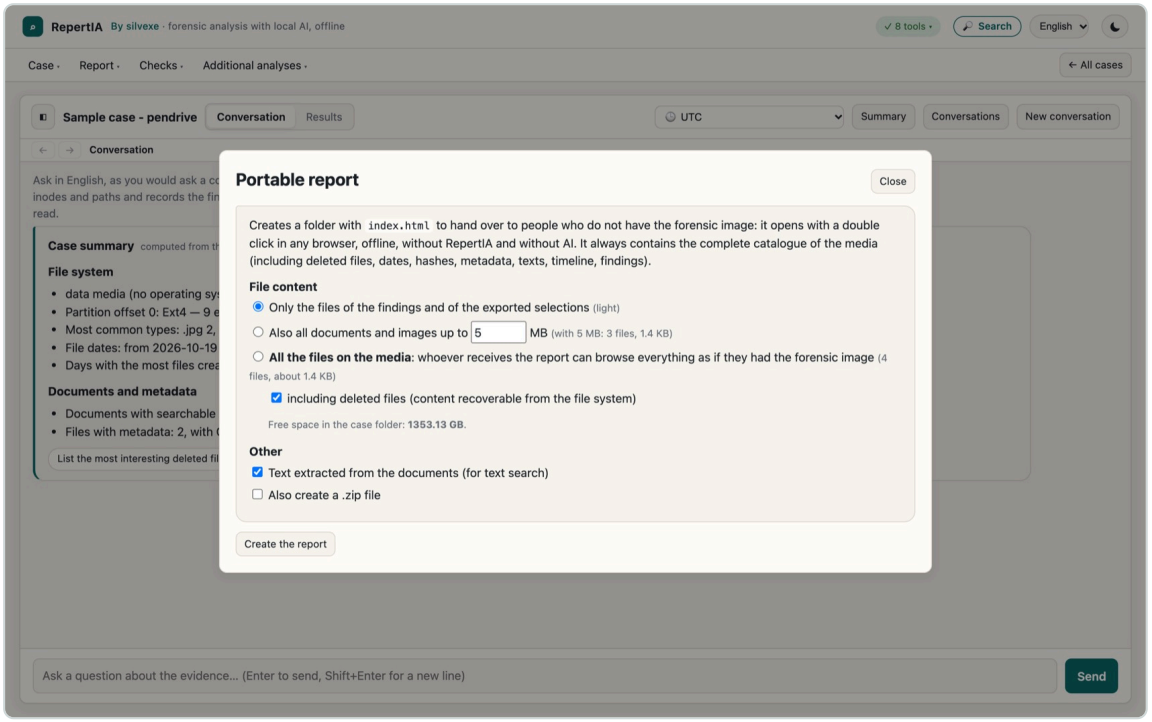


Figure 16. It creates a folder with `index.html` for people without the forensic copy: it opens with a double-click and does not need RepertIA. The window estimates the space required.

5. Smartphones

A smartphone case uses the same interface, reports and log as a disk. The examples use three test cases: an iPhone with sysdiagnose, an iPhone with a keychain and an Android phone with a WhatsApp backup.

New smartphone case

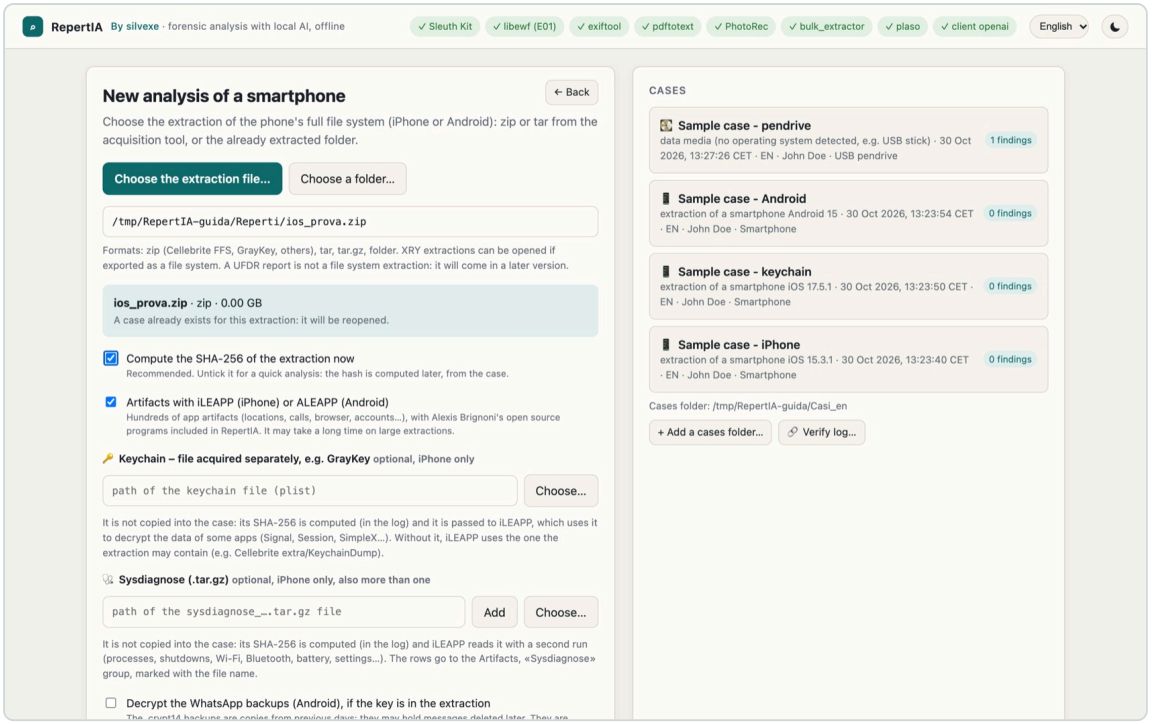


Figure 17. From the start page choose **Smartphone** and point to the zip, tar, tar.gz or folder. A tar.gz is unpacked into a derived copy inside the case; the original is not touched. The iLEAPP/ALEAPP option adds hundreds of app artifacts.

Case options

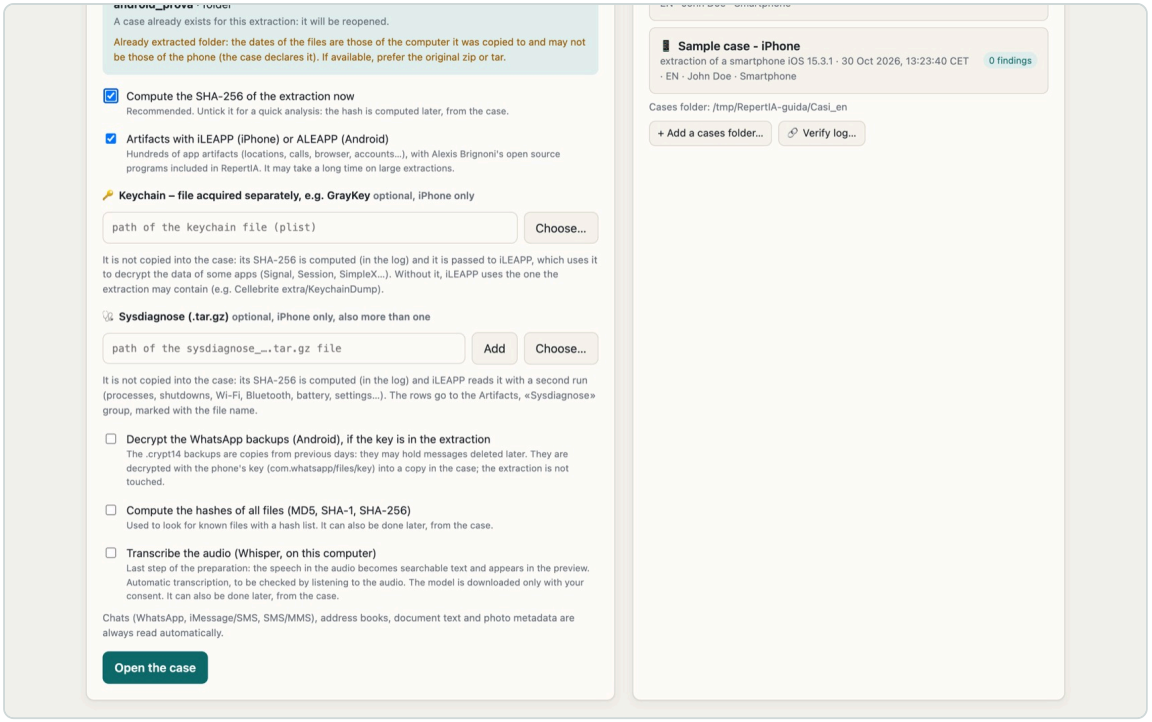


Figure 18. This is where you add the **keychain** and **sysdiagnose** archives, for iPhone only, and turn on **WhatsApp backup decryption**, for Android only. Chats, address books, text and metadata are always read.

Case summary

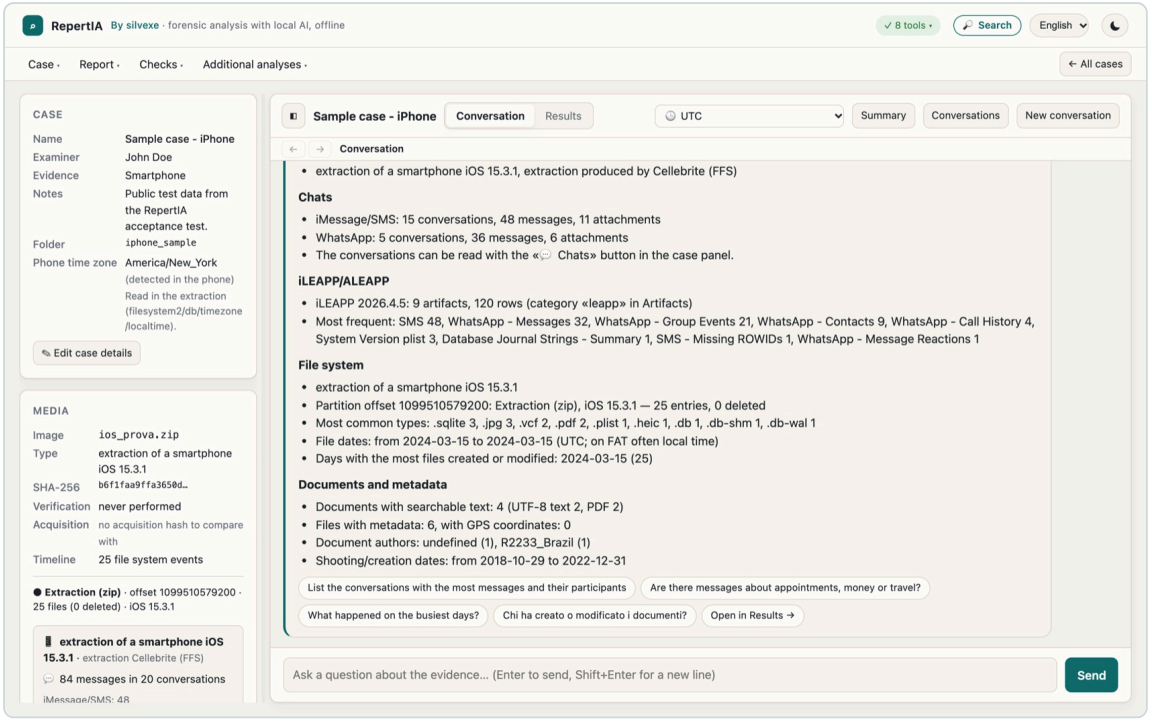


Figure 19. The summary shows chats, artifacts, files and documents. System, version and phone time zone are taken from the extraction.

Smartphone panel

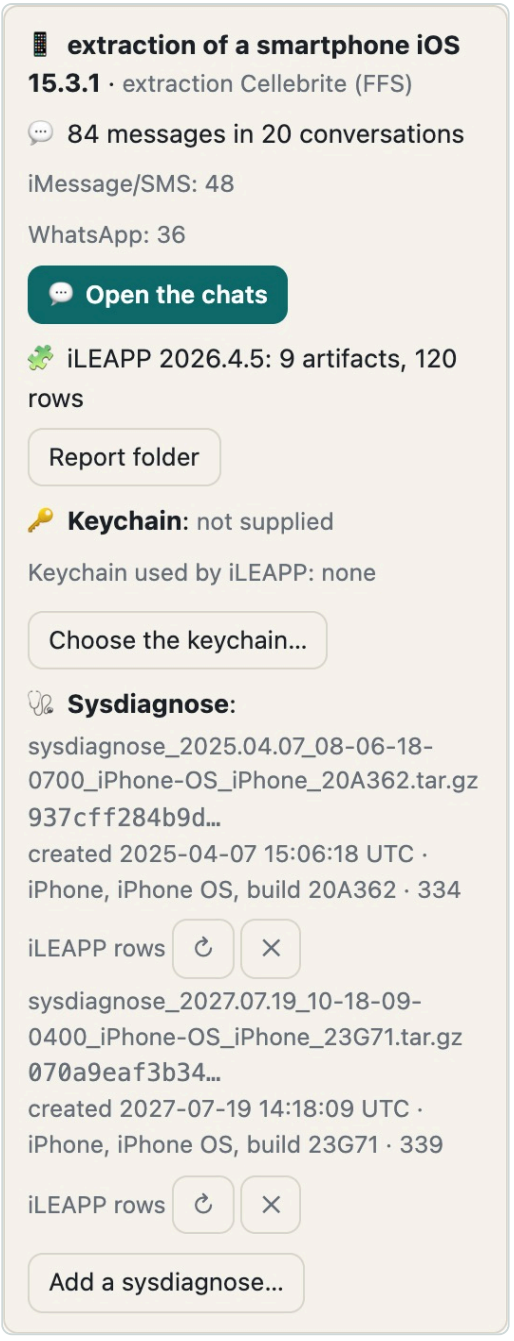


Figure 20. The case panel shows the counts, the button that opens the chats, the iLEAPP report folder, the keychain and the sysdiagnose archives, which can also be added, re-read or removed later.

Chat view

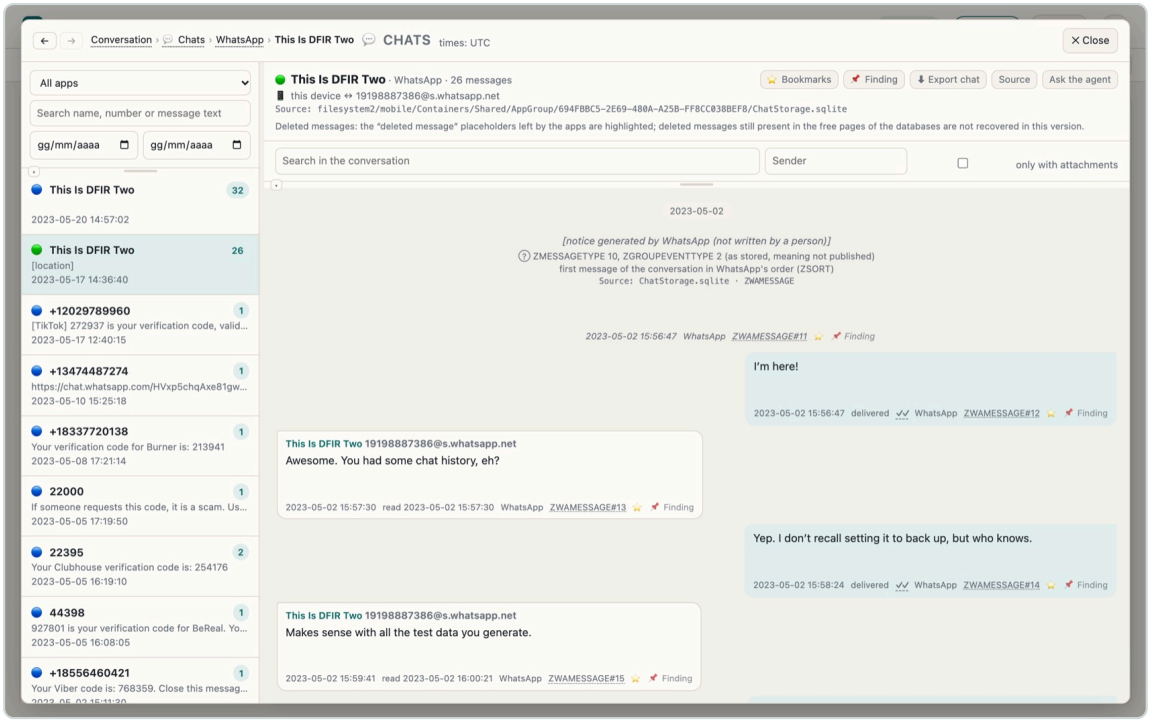


Figure 21. Conversations on the left, with filters by app, name, number, text and period. Message bubbles on the right with time, state and attachments. Each message shows its source database, table and row. **Bookmarks**, **Finding** and **Export chat** work on a single message or the whole conversation.

Calls

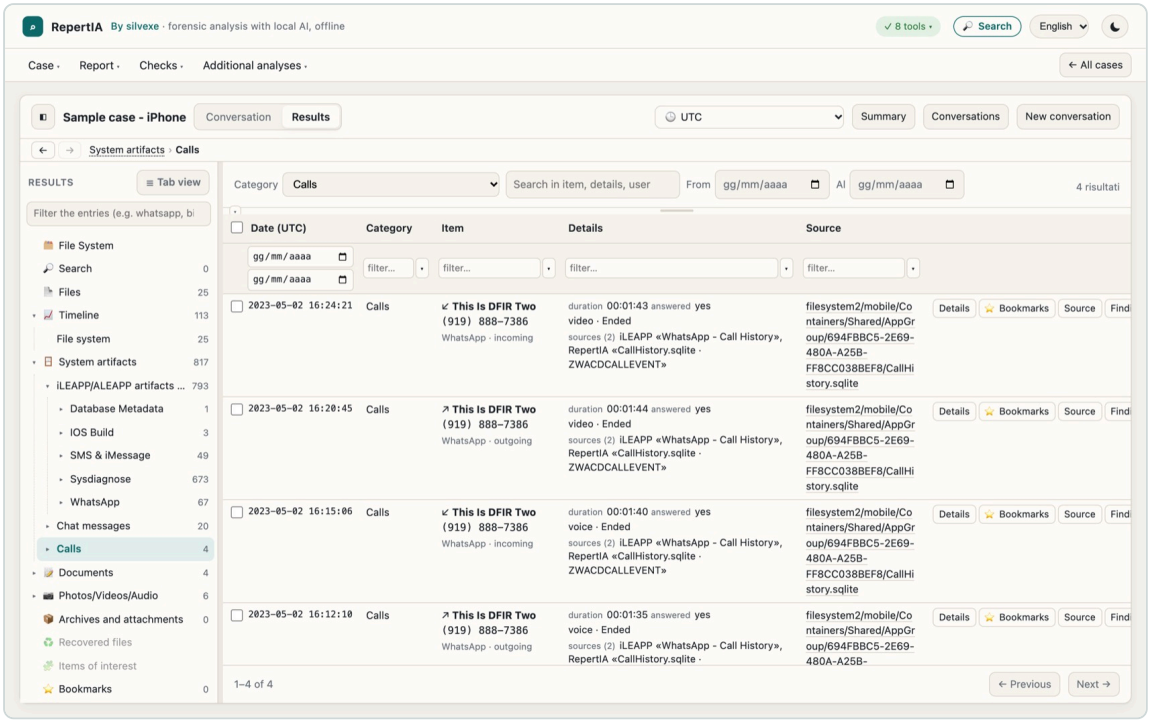


Figure 22. One row per call, with date, direction, number, name, duration, app and all sources. Identical rows from different sources are merged: here iLEAPP and RepertIA's own reader.

iLEAPP and ALEAPP artifacts

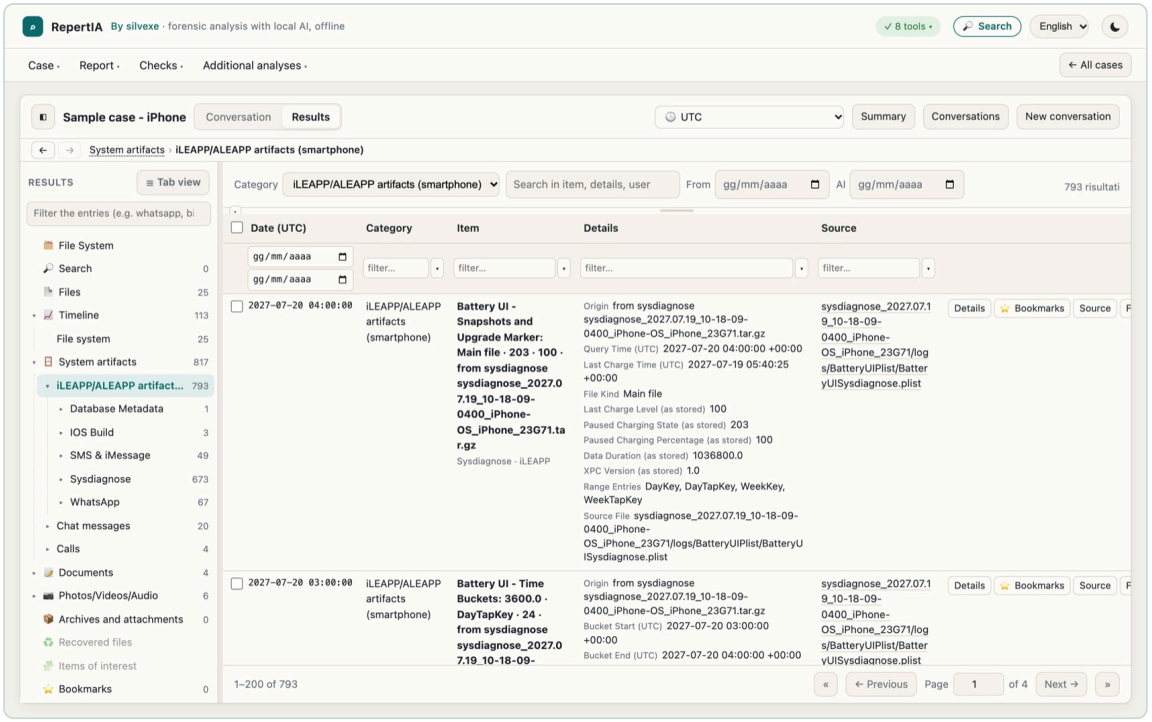


Figure 23. iLEAPP and ALEAPP tables become artifacts, split in the tree by category and artifact, with the row count. The Details column shows the fields, with dates in the chosen time zone.

Sysdiagnose

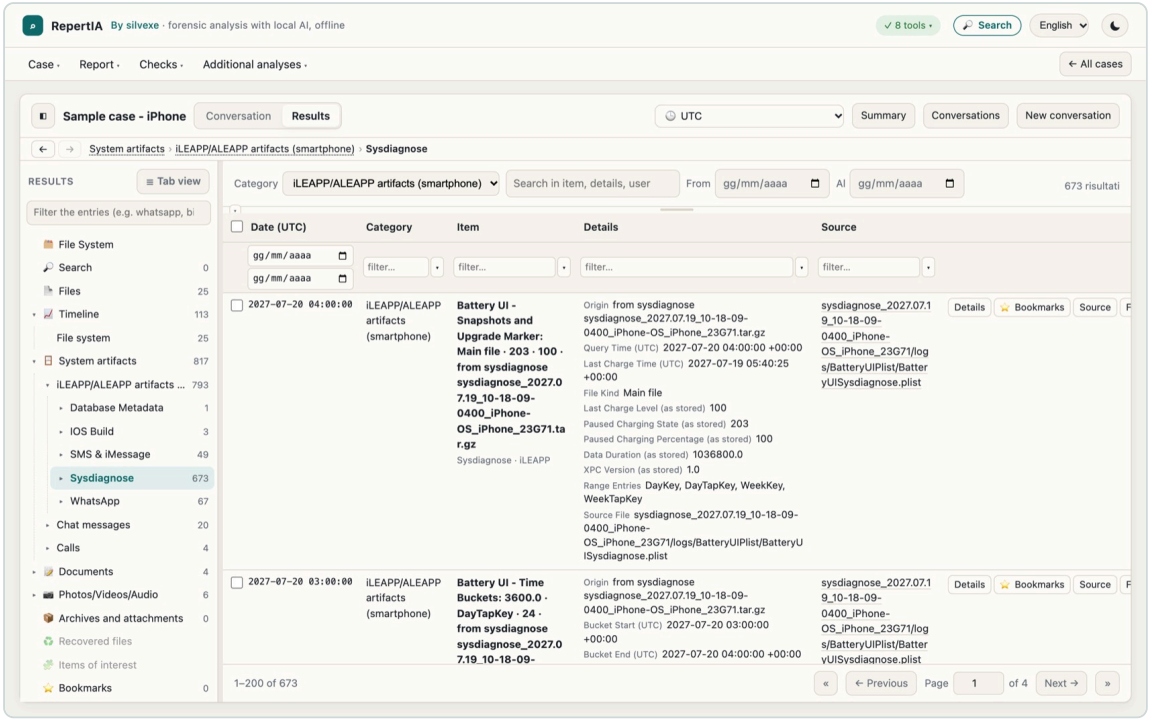


Figure 24. A sysdiagnose is the archive an iPhone produces on request: processes, shutdowns, Wi-Fi, Bluetooth, battery, settings. RepertIA records its SHA-256, does not copy it into the case and has iLEAPP read it in a second run. Rows go into the **Sysdiagnose** group, with an **Origin** field. The archives used here are built from public excerpts, not complete sysdiagnose archives.

A sysdiagnose row

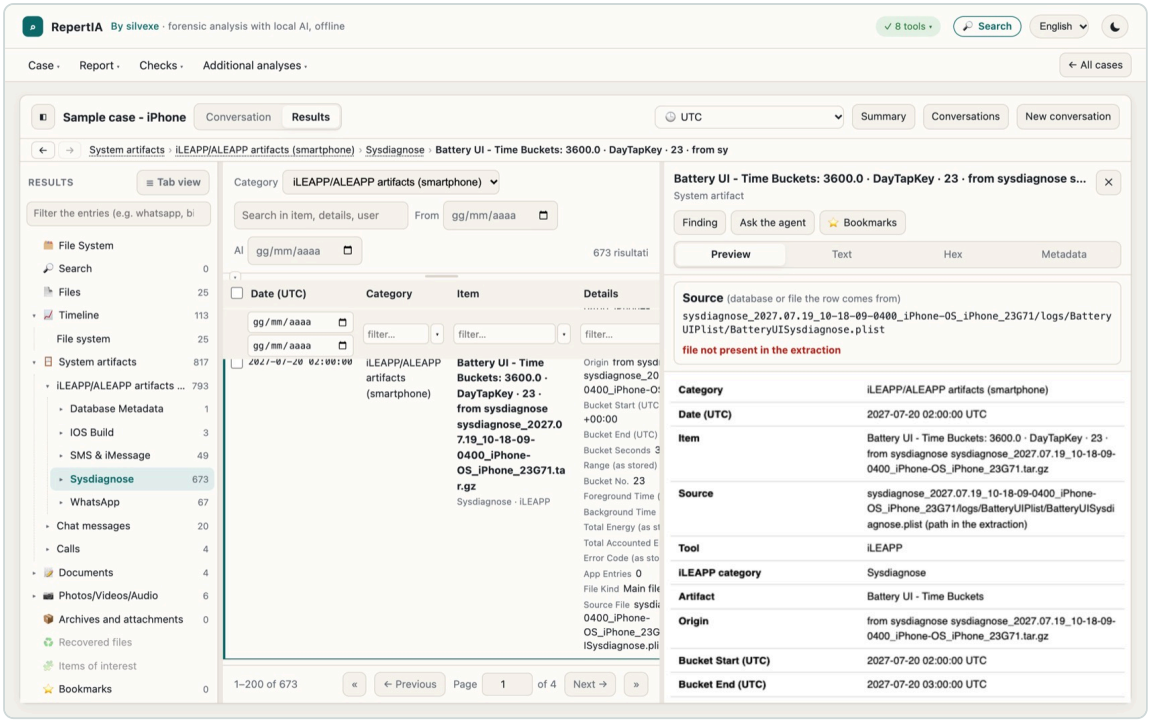


Figure 25. The detail shows the file the row comes from and warns that it is not in the extraction, because it is inside the separately acquired sysdiagnose. Creation date and device come from the archive's folder name, written by the iPhone.

Keychain and secret values

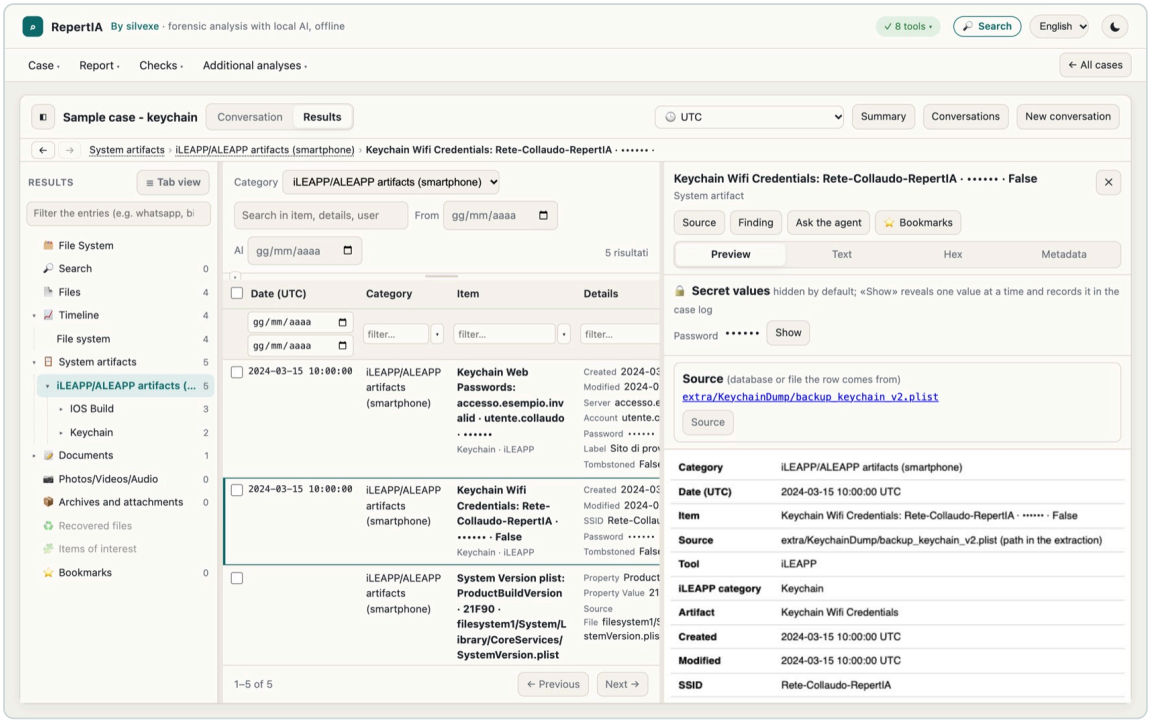


Figure 26. The iOS keychain is acquired separately. RepertIA records its SHA-256 and passes it to iLEAPP, which decrypts and reads it. Passwords, keys and tokens are hidden in tables, search, reports and from the agent. The keychain here is fake.

Showing a secret value

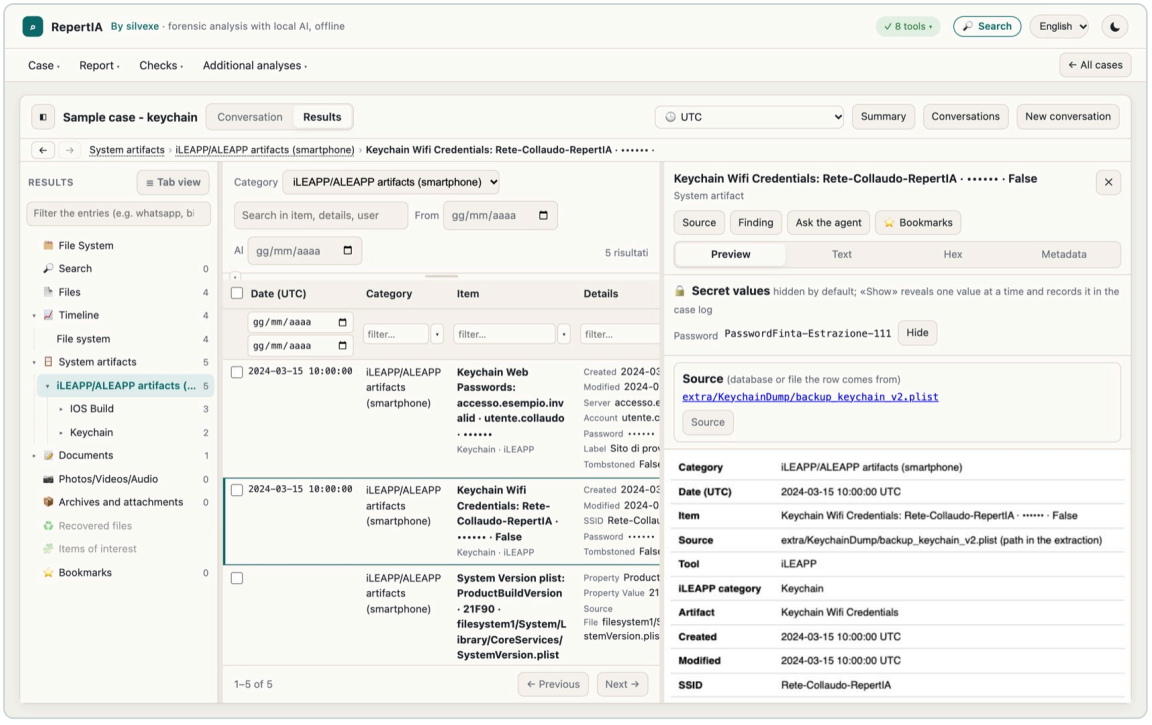


Figure 27. Show reveals one value at a time and the access goes into the case log, without the value. Secrets enter the portable report only by explicit choice. Note: the iLEAPP report in the case folder contains them in clear text.

WhatsApp backup in the case

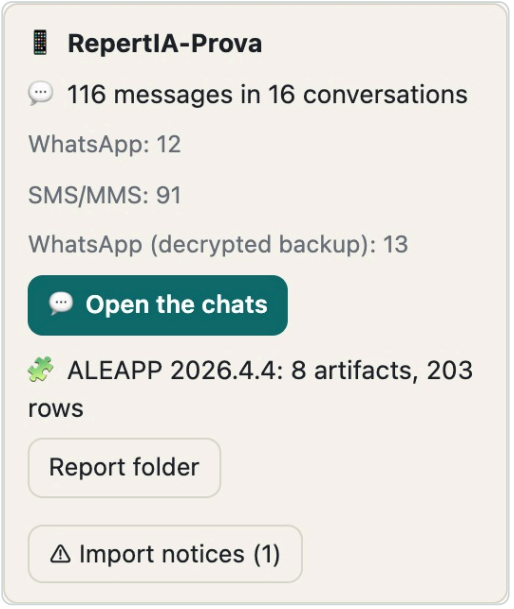


Figure 28. On Android, WhatsApp keeps encrypted `msgstore*.db.crypt14` copies. With the option on, or later from Additional analyses, RepertIA decrypts them if the key `com.whatsapp/files/key` is in the extraction. The backup appears as a separate app: here 13 messages against 12 in the current database. The backup and key in this case are fake.

Backup conversations

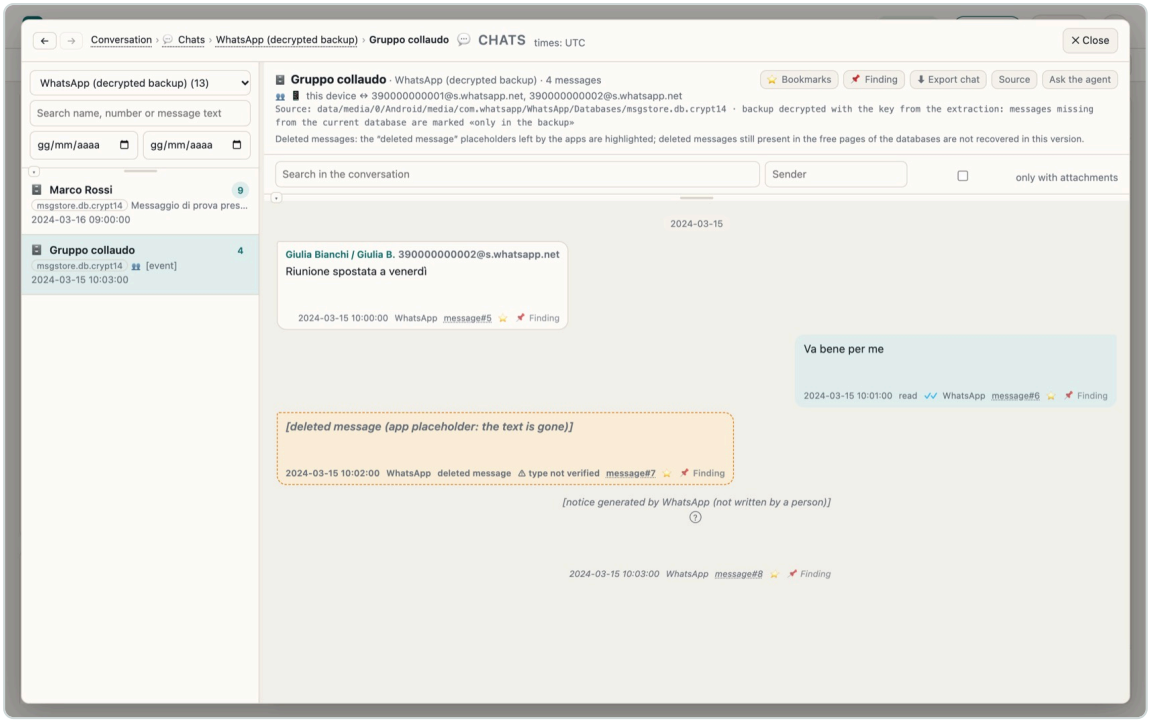


Figure 29. Filter the chats by **WhatsApp (decrypted backup)**. The header shows the source file. Messages missing from the current database are marked "only in the backup": they may have been deleted after the copy was made. Only crypt14 is handled: crypt15 keys are not on the phone, and incremental backups are only listed.

Deleted messages

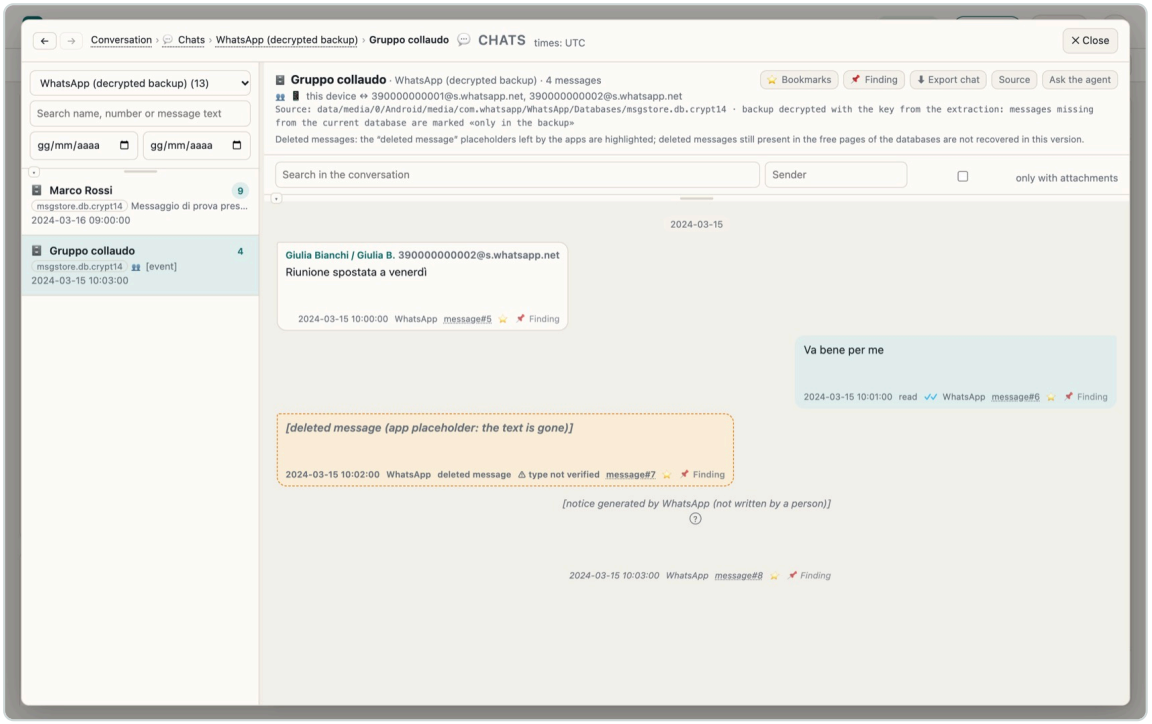


Figure 30. The "message deleted" placeholders left by the app are highlighted. The original text is gone: messages still present in free database pages are not recovered in this version.

6. The AI agent

The agent is optional. The model runs on your computer: questions and data read stay there unless you choose a network server. There are three steps: connect the model, ask questions, check the findings.

Connecting the model

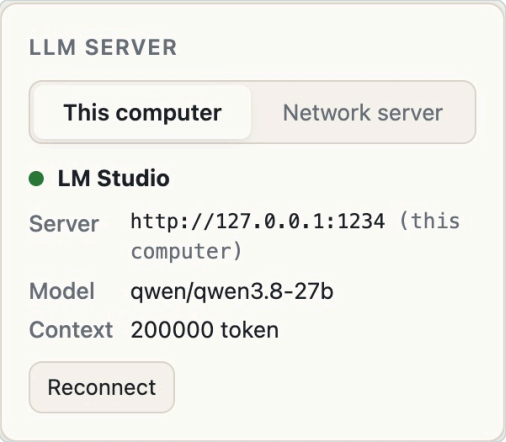


Figure 31. Start the server in LM Studio from the **Developer** tab and open the case. In the **LLM server** box, at the bottom of the left panel, with **This computer** RepertIA finds the server by itself on ports 8080, 1234 and 11434. The green dot shows the connection, with server, model and context. If you start the server after opening the case, press **Reconnect**.

A server on the network

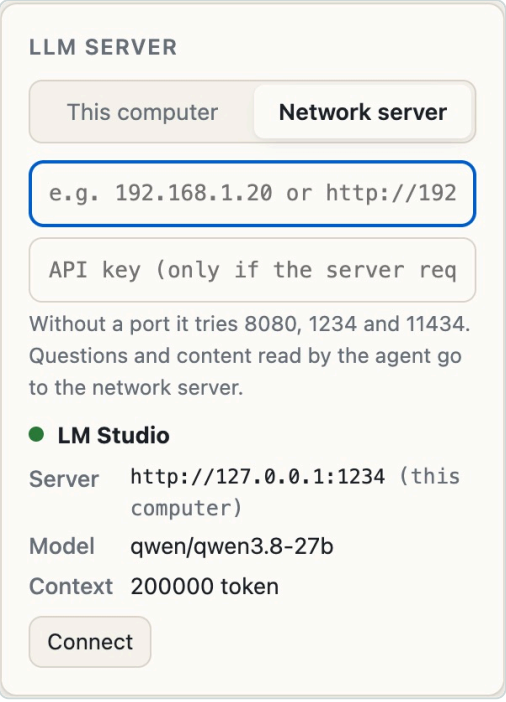


Figure 32. With **Network server** one powerful computer can serve several workstations. Enter the address, the API key if needed, and press **Connect**: without a port it tries 8080, 1234 and 11434. In LM Studio turn on *Serve on Local Network*. Questions and content read by the agent go to that server, and the report says so. If the address is outside the local network, RepertIA asks for explicit confirmation and records it.

Asking a question

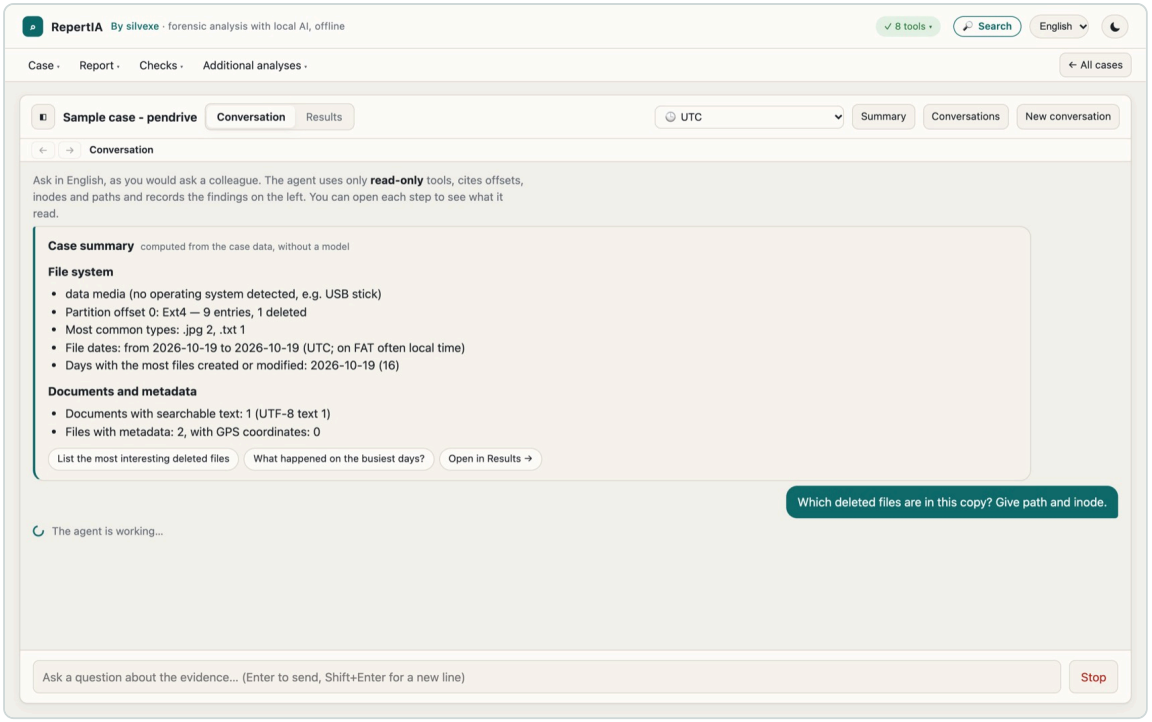


Figure 33. In the **Conversation** tab, write as you would to a colleague. The agent works only with read-only tools: each step stays visible and can be opened to see what it read.

The answer

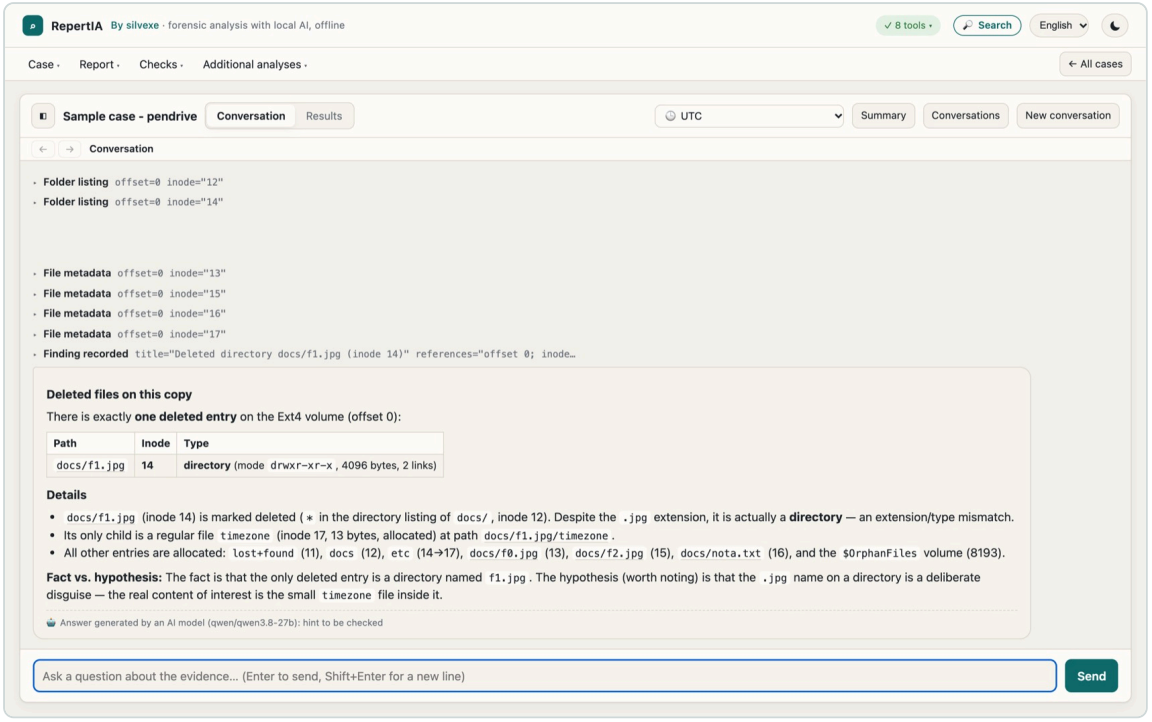


Figure 34. A real answer from the local qwen3.8-27b model in LM Studio. It separates **facts** and **hypotheses** and cites inodes and paths. Its hypothesis is wrong, though: `f1.jpg` is not a disguised folder, its inode was reassigned to `/etc`. The label at the bottom reminds you it is a lead to verify. The recorded finding stays "to be checked" until you confirm it, as shown next.

Checking and confirming a finding

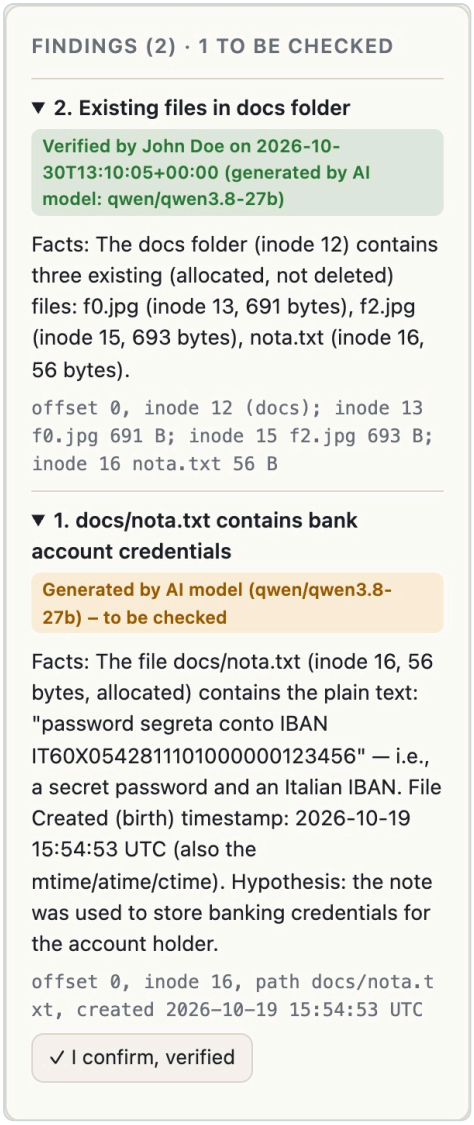


Figure 35. The **Findings** box is on the left, below LLM server: the arrow opens the text. Each model finding starts as "to be checked" and must be checked on the forensic tools, not on the answer.

No. 2 is confirmed. Files, inodes and sizes match the Results tab. With **✓ I confirm, verified** RepertIA asks for the operator's name and records name and date in the case, the log and the report.

No. 1 must not be confirmed. It is a real model error. The file was created at a single instant, written differently in each time zone. The disk is set to India, 5 hours 30 minutes ahead of UTC:

UTC	**10:24:53**
India (device)	15:54:53
Italy, October 2026	12:24:53

The model writes **15:54:53 UTC**: the number is India's, the label is wrong. The file then appears to have been created 5 hours 30 minutes later than it was. In a real case an error like this can turn a "before" into an "after", for example relative to a login or a seizure. Always compare with the Timeline, which here shows 10:24:53 UTC. Leave the finding "to be checked" and write the correct data with **Record finding**.

Answers can also change from one run to the next: in other tests the same model gave the right time and correctly explained the reassigned inode of `f1.jpg`.

7. Features described without screenshots

These features exist in the program but cannot be shown meaningfully with test data. They are described without screenshots, so as not to present constructed results.

- **Encrypted volumes.** BitLocker, LUKS1 and FileVault are decrypted into a new copy if you know the password or key; the original does not change. LUKS2 is recognised but not decrypted. APFS volumes are unlocked with a password or recovery key, kept in memory only.
- **plaso timeline.** Adds registry, prefetch, log and browser events. On large disks it can take hours.
- **Carving.** PhotoRec recovers files from free space and bulk_extractor looks for emails, sites, phone numbers and GPS data across the disk.
- **Audio transcription.** Whisper transcribes on your computer, without network access. Models are installed separately. The text is always marked as an automatic transcription to verify by listening to the audio.
- **Cross-check with another tool.** Compares results with exports from FTK Imager, Autopsy or a body file. It has not been tested with real exports from every program.
- **Multiple Windows installations, artifacts and shadow copies.** These are read from real system disks: the test image has none.

For details, formats and limits of each feature see the full manual `docs/MANUAL.md` and `VALIDATION.md` in the repository github.com/silvexe/RepertIA.